# Capstone Train, Analyze, Improve

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

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## Recall and choose your starting point

> Record your starting explanation for Capstone Train, Analyze, Improve before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Capstone Dataset and Baseline

Without opening the old lesson, explain one key idea from Capstone Dataset and Baseline. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Proxy Features and Fairness

Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Feature Engineering

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 21

## Practise with fading help

> Complete the supported Capstone Train, Analyze, Improve trace and record the first step you can explain without help.

Use W2: Audit a difference for Capstone evaluation and revision. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

## Individual reasoning check

> Check the proposed Capstone Train, Analyze, Improve explanation against the supplied facts before discussing it.

Use the worked case for Capstone evaluation and revision. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 21

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

**Independent case: Apply independently**

Frozen candidate[[6, 2],[1, 3]] and baseline[[8, 0],[4, 0]] on 12 cases. Compute candidate metrics and compare error types.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 21

## Your learning target

> Connect the Capstone Train, Analyze, Improve target to a trace and a justified conclusion.

A new model reduces validation errors from 8 to 5 but the team also removed three hard cases. Can improvement be attributed to the model? Propose a repair to the experiment.

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WORKBOOK · 6 OF 21

## Solve the given case

> Solve the supplied Capstone Train, Analyze, Improve case and show the deciding evidence.

Baseline validation accuracy**70%**; model**80%**; most remaining errors are dim images.

Plan improvement **evidence**.

Show your trace, calculation or **evidence**

**Write or draw your reasoning and evidence.**

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WORKBOOK · 7 OF 21

## Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A revision raises overall validation accuracy from **80%** to 84% but lowers dim-light accuracy from **60%** to 40%. How should the improvement claim change?

My independent answer and reason

**Write or draw your reasoning and evidence.**

**Week 35 Test and reflect**

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WORKBOOK · 8 OF 21

## Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

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## Project checkpoint

> Justify what Capstone Train, Analyze, Improve adds to Responsible Plant Classifier and keep the evidence.

Finalize the Plant Classifier experiment record, confusion matrix, subgroup table, model card, and honest remaining-limit statement.

The project change and **evidence** I will keep

**Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

## Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

## Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A new model reduces validation errors from 8 to 5 but the team also removed three hard cases. Can improvement be attributed to the model? Propose a repair to the experiment.

**My independently reasoned answer**

Record support used: none, hint or teacher explanation.

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WORKBOOK · 12 OF 21

## W1 Score the frozen test

> Score the frozen test. Apply the supplied Capstone Train, Analyze, Improve facts and keep your reasoning.

### Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Capstone Train, Analyze, Improve", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Compute the worked candidate metrics and baseline accuracy.

**My reasoning and evidence**

WORKBOOK · 13 OF 21

## W2 Audit a difference

Audit a difference. Apply the supplied Capstone Train, Analyze, Improve facts and keep your reasoning.

What does the one-correct-case improvement establish?

**My reasoning and evidence**

WORKBOOK · 14 OF 21

## W3 Label a revision

Label a revision. Apply the supplied Capstone Train, Analyze, Improve facts and keep your reasoning.

A threshold changes after test errors are read. Label its score on the same rows.

**My reasoning and evidence**

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WORKBOOK · 15 OF 21

## W4 Apply independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Capstone Train, Analyze, Improve facts and keep your reasoning.

### Rules supplied for this independent case

Matrix convention: rows are reference [clear, blocked]; columns are prediction [clear, blocked]; blocked is positive. Each matrix is [[TN, FP], [FN, TP]]. Use the same twelve reference cases for candidate and baseline.

Frozen candidate[[6, 2],[1, 3]] and baseline[[8, 0],[4, 0]] on 12 cases. Compute candidate metrics and compare error types.

**My reasoning and evidence**

WORKBOOK · 16 OF 21

## W5 Update the card

Update the card. Apply the supplied Capstone Train, Analyze, Improve facts and keep your reasoning.

What must remain visible when version 2 is proposed without a fresh test?

**My reasoning and evidence**

WORKBOOK · 17 OF 21

## Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |

| Evidence | My record |
| --- | --- |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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WORKBOOK · 18 OF 21

## Pictorial case practice: Capstone Train, Analyze, Improve

> Choose the next capstone revision from explicit false alarms and misses. Explain the labelled step before checking the changed condition.

Choose the next capstone revision from explicit false alarms and misses.

Positive=spotted. CandidateTP4, FN 2, FP 1, TN 9; baselineTP0, FN 6, FP 0, TN 10. Both evaluate the same16 cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/35

# Candidate, before revision

| | predicted spotted | predicted healthy |
|---|---|---|
| actual spotted | 4 | 2 |
| actual healthy | 1 | 9 |

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/35

## Capstone Train, Analyze, Improve

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



### 1. Compare accuracy

Candidate13/16=81.25%; baseline10/16=62.5%.



### 2. Compare detection

Candidate catches4/6 spotted leaves; baseline catches0/6.



### 3. Inspect errors

Review candidate's two misses and one false alarm with conditions and label evidence before revising.

**CHANGE ONE THING**

A revision fixes one miss but creates two extra false alarms.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is “the revision is better” incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

| Evidence to keep | My record |
|---|---|
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |

| Evidence to keep | My record |
| --- | --- |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Choose the next capstone revision from explicit false alarms and misses.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Review candidate's two misses and one false alarm with conditions and label **evidence** before revising.

1. Inspect errors

2. Compare accuracy

3. Compare detection

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Inspect errors Review candidate’s two misses and one false alarm with conditions and label **evidence** before revising. Read the picture in order.

Review candidate’s two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: A revision fixes one miss but creates two extra false alarms. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall. Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

### Repair a missing building block

Review Capstone Dataset and Baseline (ai-4/34). Goal: Define a feasible capstone question and acceptance criteria. Short practice: What provenance field prevents the crop from appearing independent?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Proxy Features and Fairness (ai-4/32). Goal: Identify possible proxy features. Short practice: What **evidence** contradicts the original “all setting information removed” claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

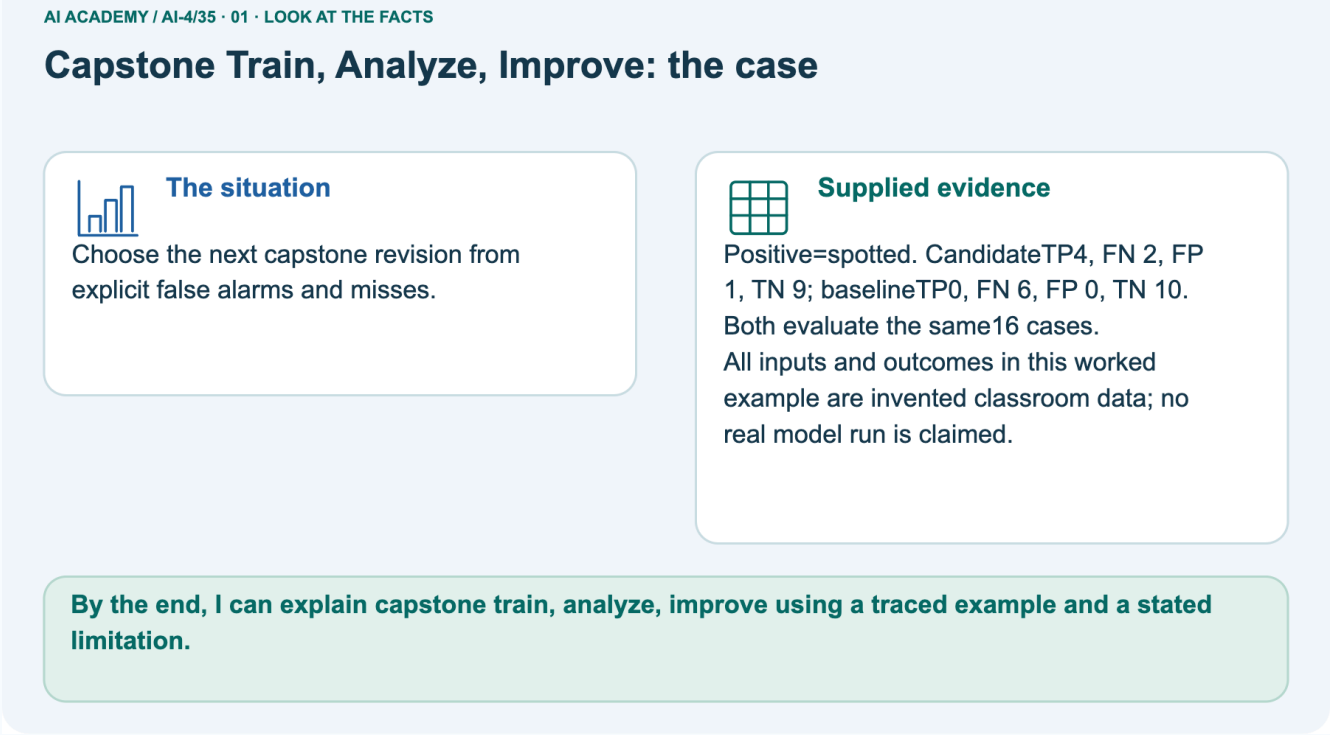
**What I repaired, the evidence I kept and the next test**

WORKBOOK · 21 OF 21

# Visual concept journey

> Explain each Capstone Train, Analyze, Improve picture from its labels, then record a relevant limit.

## 01. Capstone Train, Analyze, Improve: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

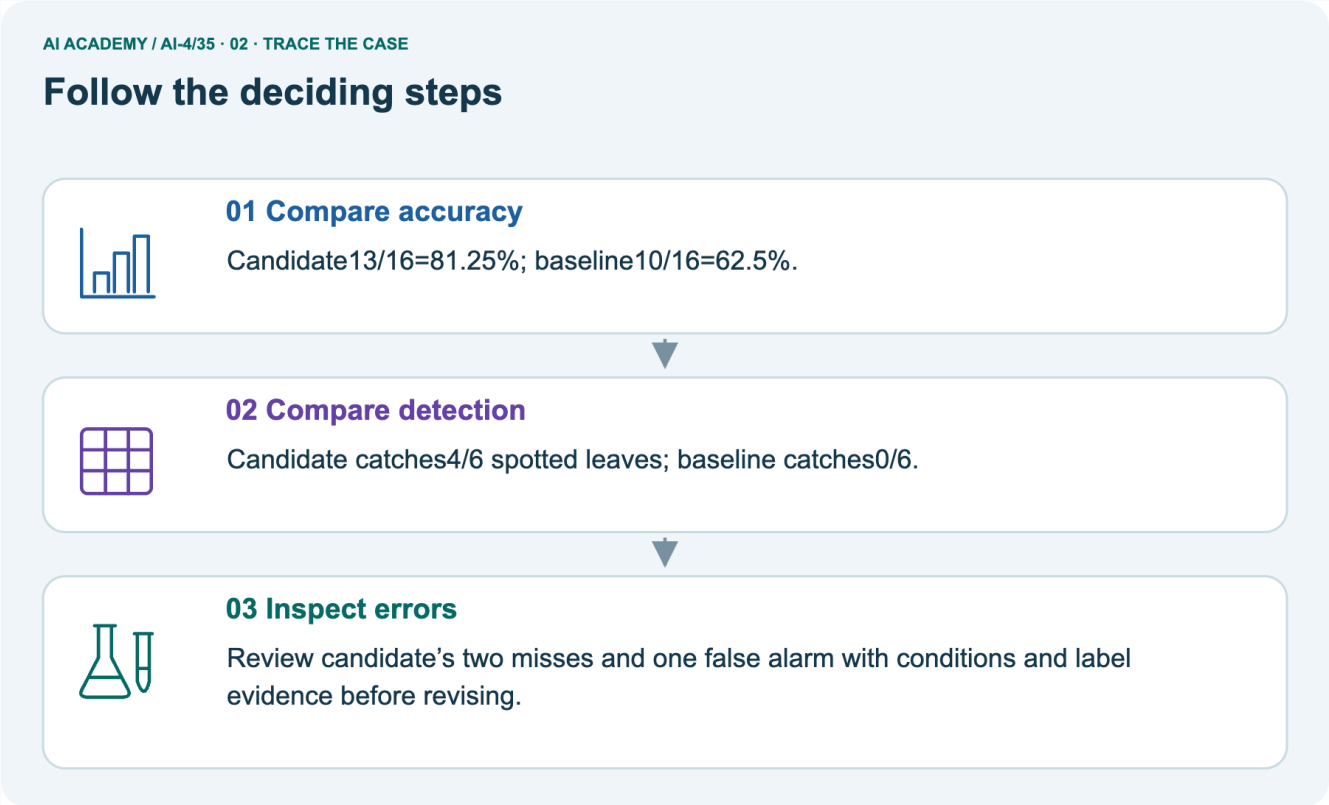
Choose the next capstone revision from explicit false alarms and misses. Supplied facts: Positive=spotted. CandidateTP4, FN 2, FP 1, TN 9; baselineTP0, FN 6, FP 0, TN 10. Both evaluate the same16 cases.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

**My picture explanation, deciding evidence and one limit**

## 02. Follow the deciding steps



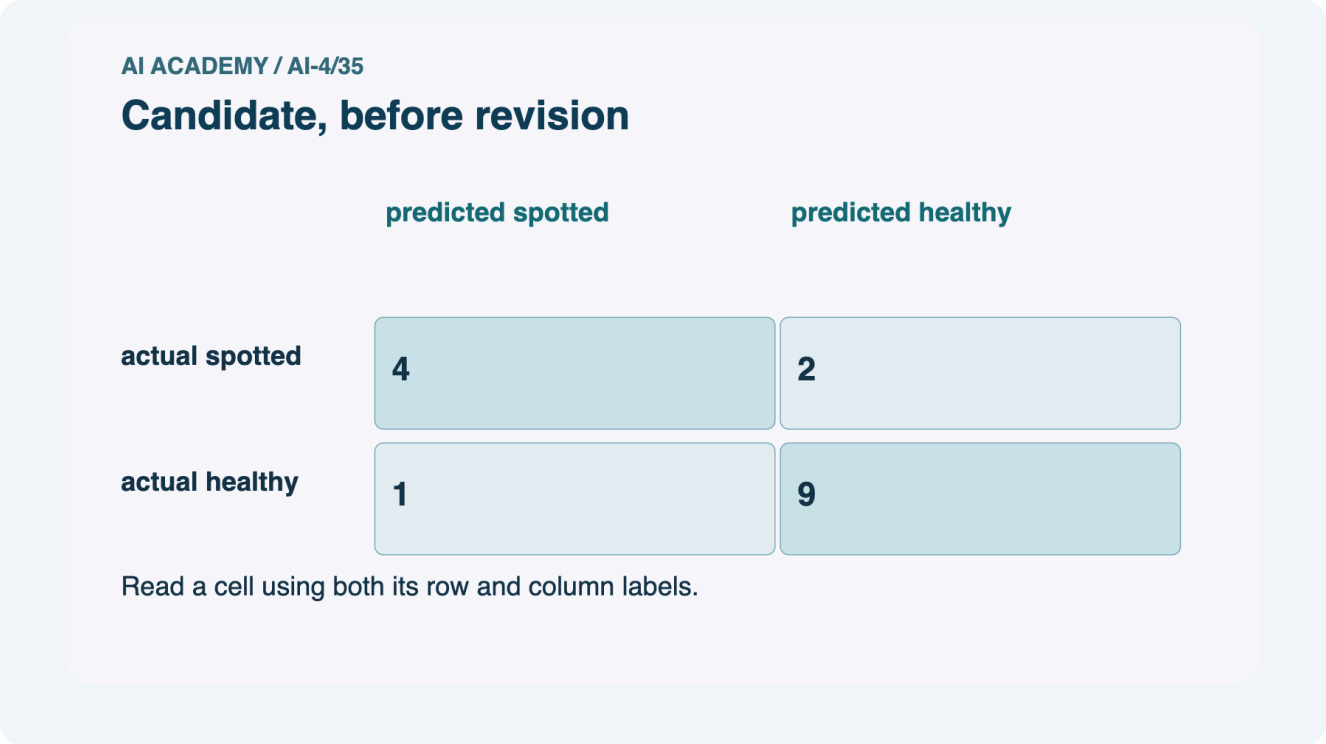

Follow the labelled picture with the matching explanation. Open the vector diagram

Compare accuracy: Candidate13/16=81.25%; baseline10/16=62.5%. → Compare detection: Candidate catches4/6 spotted leaves; baseline catches0/6. → Inspect errors: Review candidate's two misses and one false alarm with conditions and label **evidence** before revising.

> Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Candidate, before revision



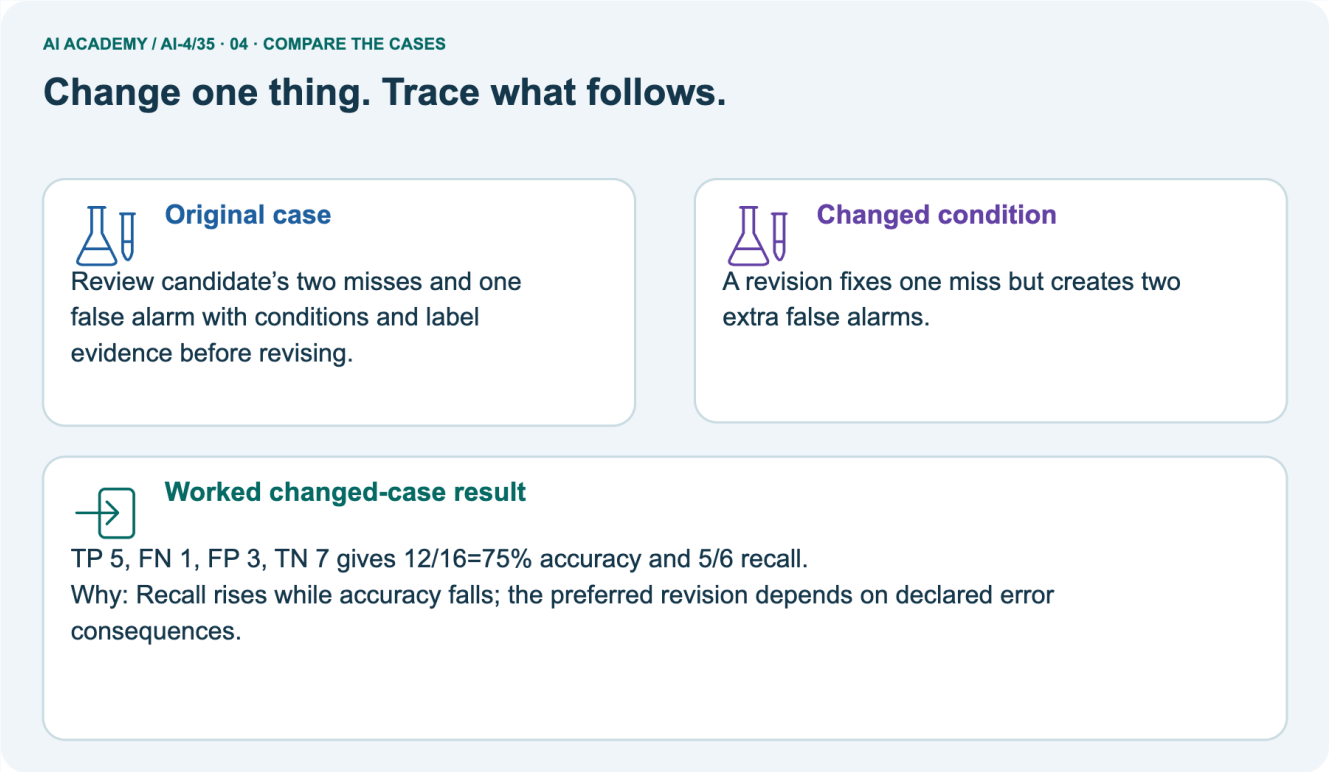

Follow the labelled picture with the matching explanation. Open the vector diagram

Candidate, before revision. actual spotted: predicted spotted = 4, predicted healthy = 2. actual healthy: predicted spotted = 1, predicted healthy = 9.

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

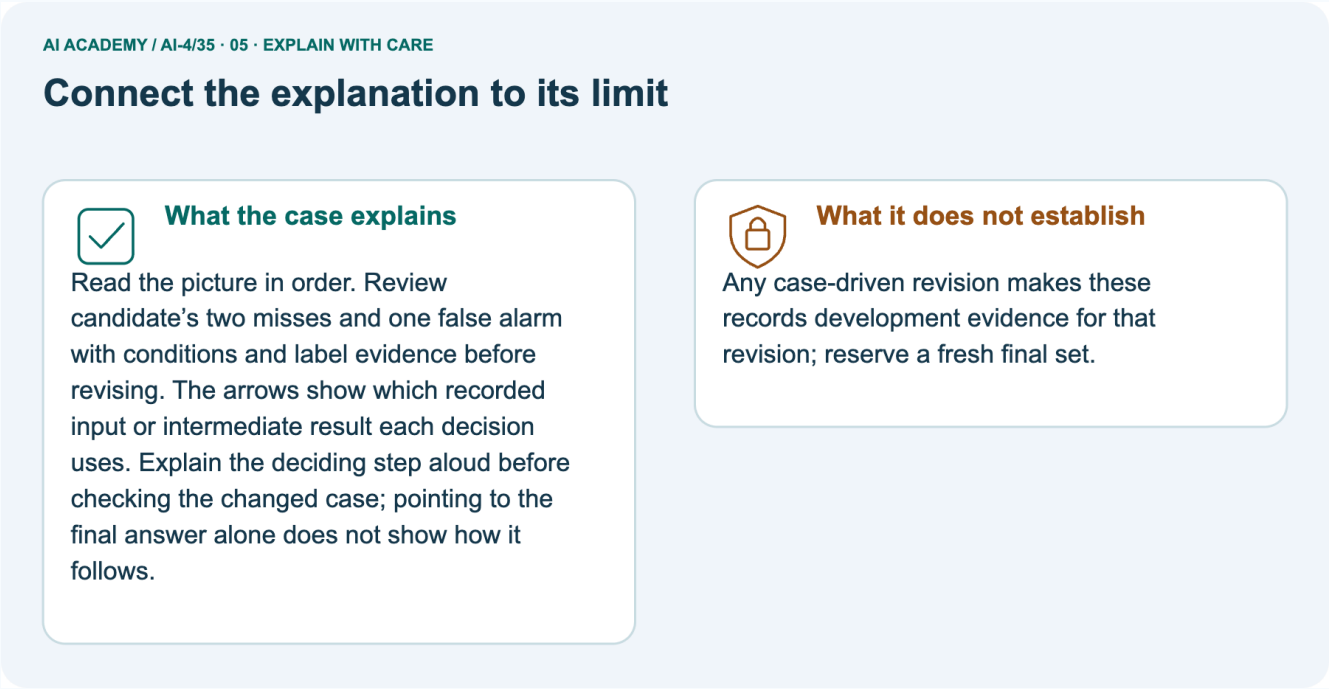
Original case: Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. Changed condition: A revision fixes one miss but creates two extra false alarms.

Expected result: TP 5, FN 1, FP 3, TN 7 gives 12/16=75% accuracy and 5/6 recall. Reason: Recall rises while accuracy falls; the preferred revision depends on declared error consequences.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Review candidate's two misses and one false alarm with conditions and label **evidence** before revising. The arrows show which recorded **input** or intermediate result each decision uses.

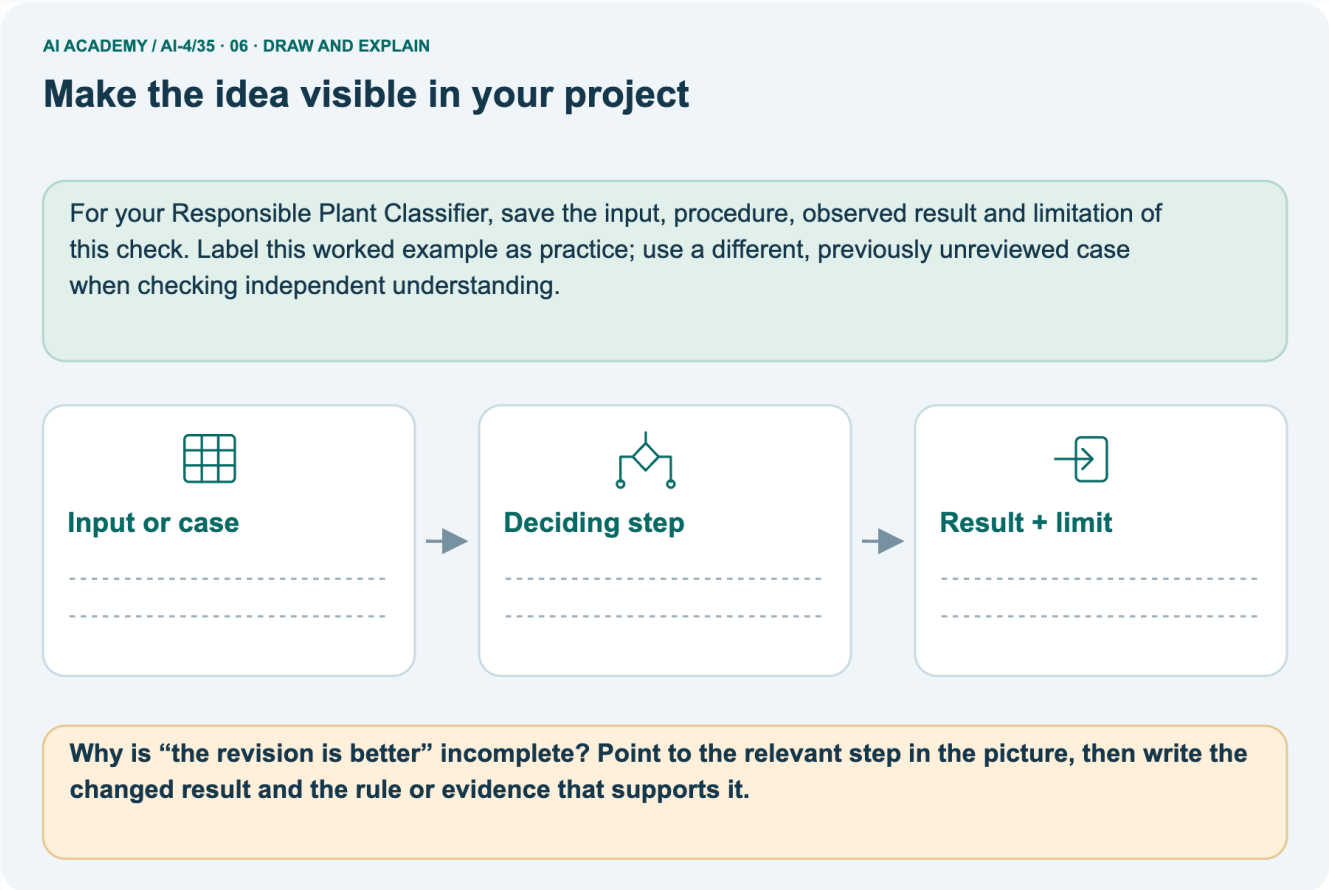
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Any case-driven revision makes these records development **evidence** for that revision; reserve a fresh final set.

Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is “the revision is better” incomplete? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.